WEEK 8 – STUDENT PERFORMANCE DASHBOARD

[Urekha V-E25DA043]

1. INTRODUCTION

The Week 8 project focuses on creating an interactive Student Performance Dashboard using Power BI. The dashboard was developed using the cleaned and transformed Student Performance in Exams dataset prepared during Week 7. It presents student performance through KPI cards, charts and interactive slicers.

2. OBJECTIVE

The main objectives of this project are:

- To create an interactive Power BI dashboard.
- To analyse student performance in different subjects.
- To display important performance indicators using KPI cards.
- To analyse Performance Grade and Pass/Fail results.
- To use slicers for interactive data filtering.
- To present the analysis in a simple and meaningful visual format.

3. DATASET

The project uses the Student Performance in Exams dataset. It contains information about students, their background and their scores in Math, Reading and Writing.

The main attributes used include Gender, Race/Ethnicity, Parental Education, Lunch, Test Preparation Course, Math Score, Reading Score and Writing Score.

The dataset was cleaned and transformed during Week 7 before being used for dashboard creation.

4. DASHBOARD DEVELOPMENT

The cleaned dataset was imported into Power BI and used to create an interactive dashboard.

The dashboard contains five KPI cards:

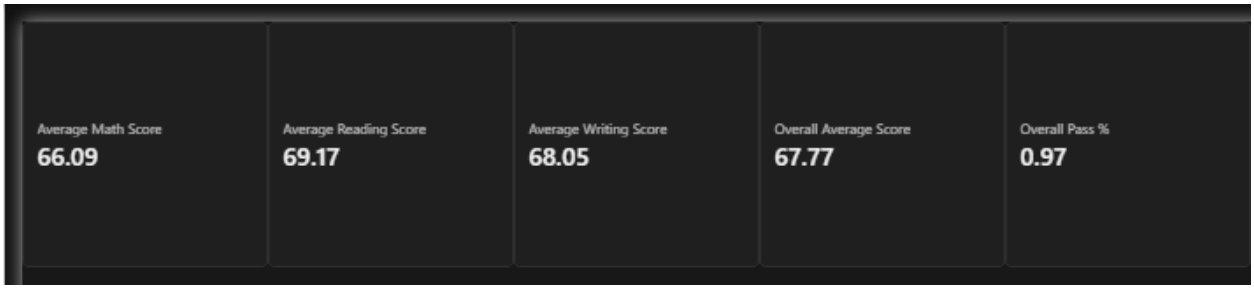
• Average Math Score

• Average Reading Score

• Average Writing Score

• Overall Average Score

• Overall Pass Percentage



Charts were also created to analyse average scores by subject, Performance Grade and Pass/Fail distribution.

5. INTERACTIVE SLICERS

Slicers were added to make the dashboard interactive. The dashboard contains slicers for:

• Gender

• Lunch

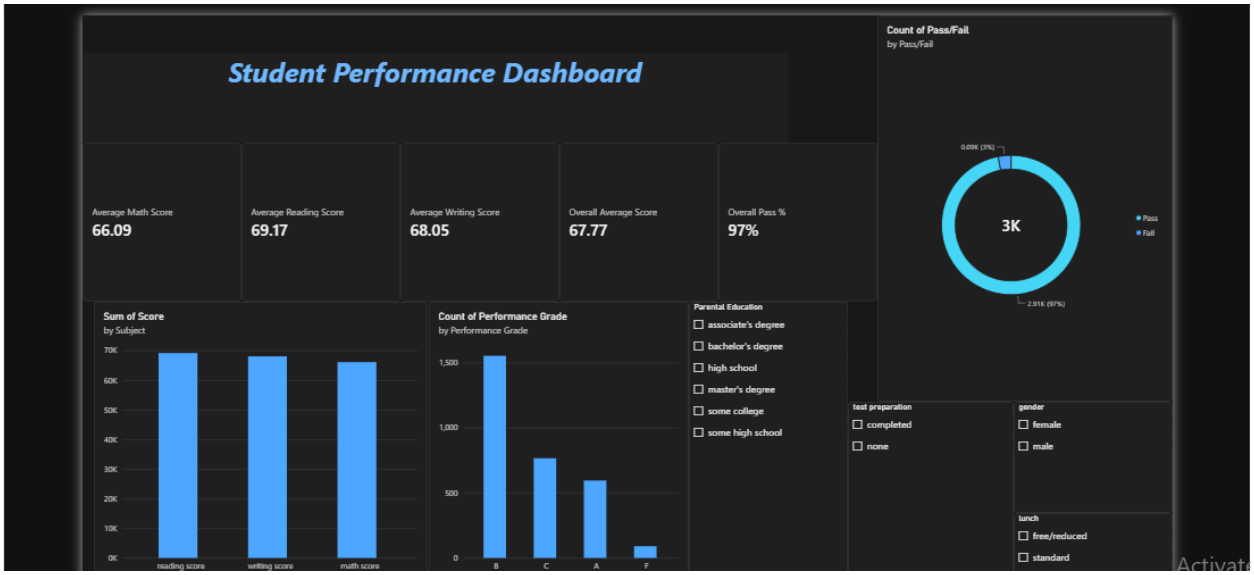
• Test Preparation Course

• Parental Education

These slicers allow users to filter the dashboard and analyse student performance based on different categories.

6. FINAL DASHBOARD

The final dashboard provides a clear visual representation of student performance using KPI cards, charts and interactive slicers.



7. RESULTS AND OBSERVATIONS

The dashboard provides a quick overview of student performance. The subject-wise analysis makes it easy to compare Math, Reading and Writing scores. The Performance Grade and Pass/Fail visuals help understand the distribution of student results.

The overall average score is approximately 65, and the overall pass percentage is approximately 96%. The slicers also allow the results to be analysed based on different student categories.

8. TOOLS USED

- Power BI Desktop
- Power Query
- DAX
- Student Performance in Exams Dataset

9. KEY LEARNING

Through this project, I learned how to create an interactive Power BI dashboard using KPI cards, charts and slicers. I also learned how DAX measures can be used to calculate and display useful performance indicators and how visualizations can be used to present data clearly.

10. CONCLUSION

The Week 8 project successfully converted the cleaned Student Performance dataset into an interactive Power BI dashboard. The dashboard provides meaningful insights into student performance and makes the data easier to analyse using visualizations and filters. This project improved my understanding of Power BI dashboard creation and data visualization.